Transformimet Furie

$$x(t) \longrightarrow X(\omega)$$

$$X(\omega) \longrightarrow \int x(t)\, e^{-j\omega t}\, dt$$

Vetit e tr. Furie

1° Linearitet

$$a_1 x_1(t) + a_2 x_2(t) \longrightarrow a_1 X_1(\omega) + a_2 X_2(\omega)$$

2° Lëvizja në kohë

$$x(t - t_0) \longrightarrow e^{-j\omega t_0} X(\omega)$$

3° Shumëzimi me $e^{j\omega_0 t}$

$$e^{j\omega_0 t} x(t) \longrightarrow X(\omega - \omega_0)$$

4° Shkallëzimi në kohë

$$x(at) \longrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

5° Pasqyrimi

$$x(-t) \longrightarrow X(-\omega)$$

6° Diferencimi në kohë

$$\frac{dx(t)}{dt} \longrightarrow j\omega X(\omega)$$

7° Shumzimi me "t"

$$t\,x(t) \longrightarrow \frac{dX(\omega)}{d\omega}$$

8° Dualiteti

$$x(t) \longrightarrow X(\omega)$$

$$X(t) \longrightarrow 2\pi x(-\omega)$$

9. Thurrja

$$x_1(t) * x_2(t) \longrightarrow X_1(\omega)\,X_2(\omega)$$

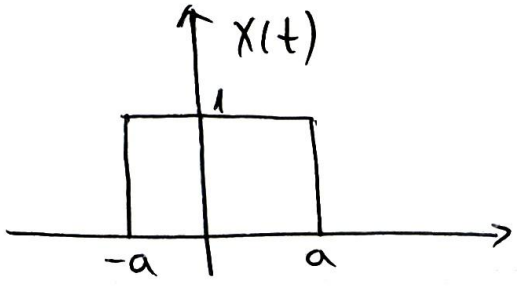
10. Shumzimi në kohë

$$x_1(t) \cdot x_2(t) \longrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

5.19

Gjeni tr. furie të sinjalit drejtkëndësh $x(t)$ në fig i cili është dhënë:

$$x(t) = P_a(t) = \begin{cases} 1 & |t| < a \\ 0 & |t| > a \end{cases}$$



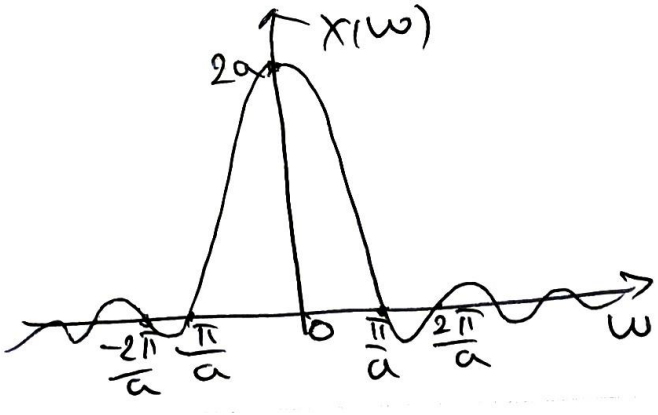
$$X(t) = P_a(t) = \begin{cases} 1 & |t| < a \\ 0 & |t| > a \end{cases}$$

$$X(\omega) = \int_{-\infty}^{\infty} P_a(t) e^{-j\omega t} dt = \int_{-a}^{a} 1 \cdot e^{-j\omega t} dt$$

$$= -\frac{1}{j\omega} e^{-j\omega t} \Big|_{-a}^{a} = \frac{1}{j\omega} \left(e^{-j\omega a} - e^{j\omega a} \right)$$

$$= \frac{1}{j\omega} \left(e^{j\omega a} - e^{-j\omega a} \right) = 2 \frac{\sin \omega a}{\omega} = 2a \frac{\sin \omega a}{\omega a}$$

f-oni sinc



Për $\omega = 0$

$X(\omega) = 2a$

Për $\omega = \frac{\pi}{a}$

$X(\omega) = 0$

5.20.

Gjeni tr. furie të sinjalit

$$x(t) = \frac{\sin at}{\pi t}$$

Nisemi nga fakti që:

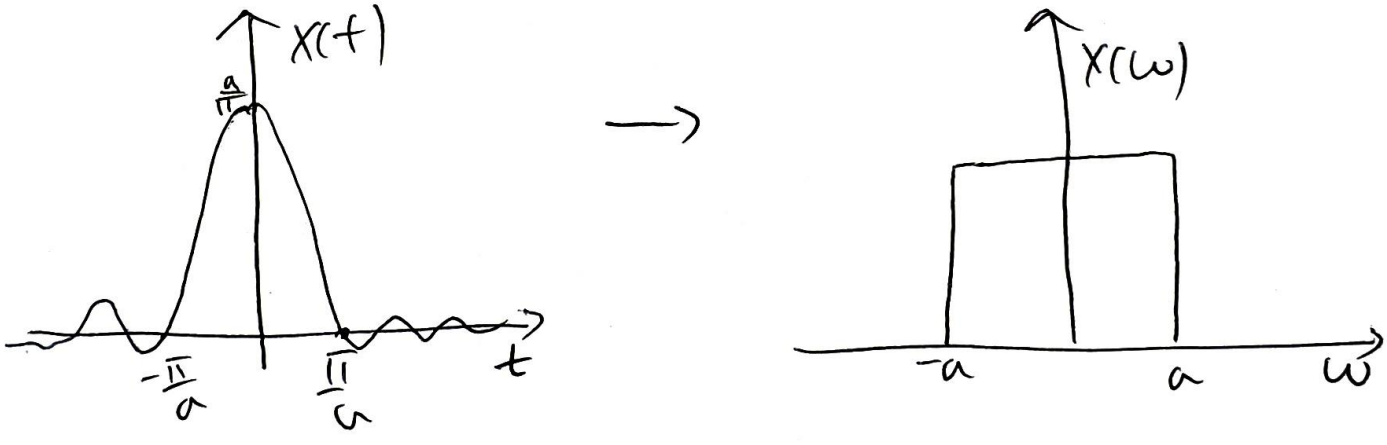
$$Pa(t) \longrightarrow 2\frac{\sin \omega a}{\omega}$$

Nga vetia e dualitetit:

$$Pa(t) \longrightarrow 2\frac{\sin \omega a}{\omega}$$

$$\frac{2\sin at}{t} \longrightarrow 2\pi\, Pa(-\omega) \;/:\pi$$

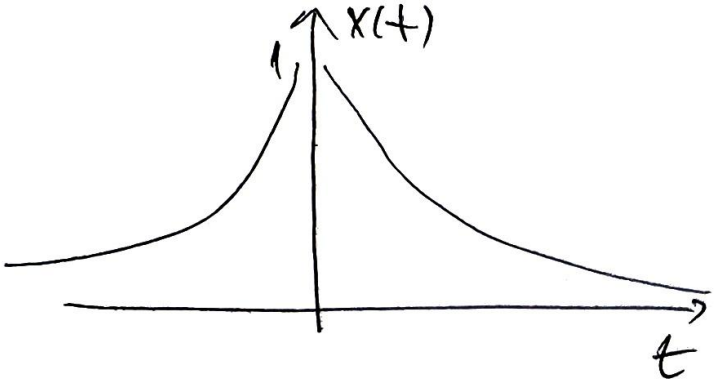
$$\frac{\sin at}{\pi t} = Pa(-\omega) = Pa(\omega)$$



5.21

Gjeni tr. furie të sinjalit:

$$x(t) = e^{-a|t|}$$



Zgjidhje

$$x(t) = e^{-a|t|} = \begin{cases} e^{-at} & t>0 \\ e^{at} & t<0 \end{cases}$$

$$X(\omega) = \int_{-\infty}^{0} e^{at} e^{-j\omega t}\, dt + \int_{0}^{\infty} e^{-at} e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{0} e^{(a-j\omega)t}\, dt + \int_{0}^{\infty} e^{-(a+j\omega)t}\, dt$$

$$= \frac{1}{a-j\omega} e^{(a-j\omega)t}\Big|_{-\infty}^{0} - \frac{1}{a+j\omega} e^{-(a+j\omega)t}\Big|_{0}^{\infty}$$

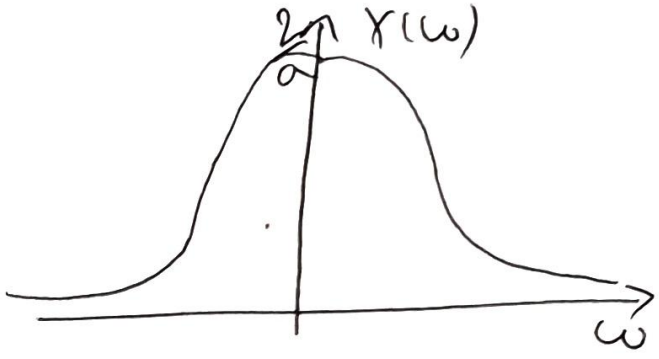
Vathdim 5.21

$$= \frac{1}{a-j\omega}(1-0) - \frac{1}{j\omega+a}(0-1)$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2+\omega^2}$$

Sqarim: $j^2 = (-1)$

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+\omega^2}$$



5.22

Gjeni tr. furie për sinjalin:

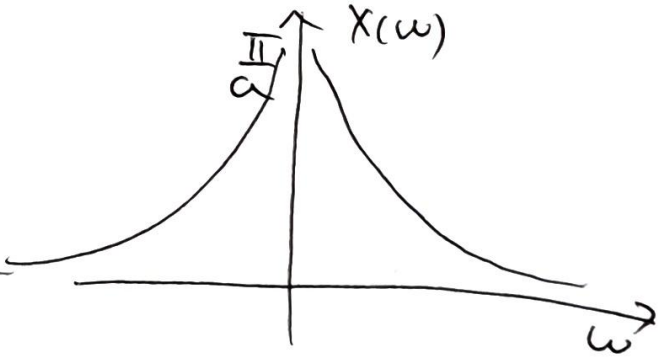
$$x(t) = \frac{1}{a^2+t^2}$$

Nisemi nga fakti që

$$e^{-a|t|} \longrightarrow \frac{2a}{a^2+\omega^2} \longrightarrow$$ Vetia e Dualitetit

$$\frac{2a}{a^2+t^2} = 2\pi e^{-a|-\omega|} = 2\pi e^{-a|\omega|} \;/\cdot a$$

$$\frac{1}{a^2+t^2} = \frac{\pi}{a} e^{-a|\omega|}$$



5.23

Gjeni tr. furie të sinjaleve

a) $x(t)=1$

b) $x(t)=e^{j\omega_0 t}$

c) $x(t)=e^{-j\omega_0 t}$

d) $x(t)=\cos\omega_0 t$

e) $x(t)=\sin\omega_0 t$

Zgjidhje

a)

$$\delta(t) \longrightarrow 1$$

$$1 \longrightarrow 2\pi\delta(-\omega)=2\pi\delta(\omega)$$

b) $x(t)=e^{j\omega_0 t}$

Nga vetia shumëzimi me $e^{j\omega_0 t}$ kemi:

$$e^{j\omega_0 t}\, x(t) = X(\omega-\omega_0)$$

$$e^{j\omega_0 t}\cdot 1 = 2\pi\delta(\omega-\omega_0)$$

c) $e^{-j\omega_0 t} = 2\pi\delta(\omega+\omega_0)$ ← rasti i njejt sikur në b) vetem ka shenjen "−"

d) $x(t) = \cos \omega_0 t$

$\cos \omega_0 t = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$ → formula e Eulerit.

$\cos \omega_0 t = \frac{2\pi}{2}\delta(\omega - \omega_0) + \frac{2\pi}{2}\delta(\omega + \omega_0)$

e) $x(t) = \sin \omega_0 (t)$

$\sin \omega_0 (t) = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right)$

$= \frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$

$= -j\pi\left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right]$

Përgjigjja Frekuencore

$$y(t) = x(t) * h(t)$$

$$Y(\omega) = X(\omega) \cdot H(\omega) \Longrightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

$H(\omega)$ ↓ përgjigjja frekuencore

5.44.

Duke përdorur tr. Furie gjeni përgjigjen frekuencore të sistemit:

$$y'(t) = 2y(t) = x(t) + x'(t)$$

$$\boxed{\frac{dx(t)}{dt} = j\omega X(\omega)}$$

$$j\omega Y(\omega) + 2Y(\omega) = X(\omega) + j\omega X(\omega)$$

$$Y(\omega)[j\omega + 2] = X(\omega)[1 + j\omega] \;/: X(\omega)$$

$$\frac{Y(\omega)}{X(\omega)} = \frac{1 + j\omega}{2 + j\omega} = \frac{2 + j\omega - 1}{2 + j\omega}$$

$$= \frac{2 + j\omega}{2 + j\omega} - \frac{1}{2 + j\omega} = 1 - \frac{1}{2 + j\omega}$$

$$\boxed{h(t) = \delta(t) - e^{-2t} u(t)}$$ → kthehemi në kohë të vazhdueshme.

5.45

Eshtë dhënë sistemi LTI di më poshtë duke përdorr

fr. fuqie gjeni daljen e sistemit.

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

a) $x(t) = e^{-t}u(t)$

b) $x(t) = u(t)$

Zgjidhje

a)

$$j\omega Y(\omega) + 2Y(\omega) = X(\omega)$$

$$Y(\omega)(j\omega+2) = X(\omega) \quad /: X(\omega) / : (j\omega+2)$$

$$\frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega+2}$$

Megense: $x(t) = e^{-t}u(t)$

$$X(\omega) = \frac{1}{1+j\omega}$$

$$Y(\omega) = X(\omega)\cdot H(\omega) = \frac{1}{1+j\omega}\cdot\frac{1}{2+j\omega} = \frac{A}{1+j\omega} + \frac{B}{2+j\omega}$$

$j\omega = -1$	$j\omega = -2$	$A = \frac{1}{1} = 1$
$A = 2 + j\omega$	$B = 1 + j\omega$	$B = \frac{-1}{1} = -1$
$A = 2 - 1 = 1$	$B = 1 - 2 = -1$	

Vathdim 5.45/

$$Y(\omega) = \frac{A}{1+j\omega} + \frac{B}{2+j\omega} = \frac{1}{1+j\omega} - \frac{1}{2+j\omega}$$

$$\boxed{y(t) = \left(e^{-t} - e^{-2t}\right) u(t)}$$

b)

$$X(\omega) = \pi\delta(\omega) + \frac{1}{j\omega} \longrightarrow x(t) = u(t)$$

$H(\omega) \rightarrow$ është c njejta si në a)

$$Y(\omega) = X(\omega) \cdot H(\omega) = \left[\pi\delta(\omega) + \frac{1}{j\omega}\right] \cdot \frac{1}{2+j\omega}$$

$$= \pi\delta(\omega)\frac{1}{2+j\omega} + \frac{1}{j\omega(2+j\omega)}$$

$$= \pi\delta(\omega)\frac{A}{(2+j\omega)} + \frac{B}{j\omega} + \frac{C}{(2+j\omega)}$$

$$\downarrow$$

$$= \frac{\pi\delta(\omega)}{2} + \frac{1}{2j\omega} - \frac{1}{2}\frac{1}{(2+j\omega)}$$

$$\frac{1}{j\omega(2+j\omega)} = \frac{A}{j\omega} + \frac{B}{2+j\omega}$$

Për A
$j\omega = 0 \Rightarrow \frac{1}{2+0} = \frac{1}{2}$

Për B
$j\omega = -2 \Rightarrow -\frac{1}{2}$